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DEEPFAKE PHISHING: HOW AI-POWERED CONTENT IS RESHAPING SOCIAL ENGINEERING

BY ABHIJEET

DISCLAIMER

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WHOAMI

Security Researcher & Tinkerer

Research areas include Adversarial TTPs, Linux systems, cloud infrastructure, and supply chain

Full-time human, part-time homechef

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Social Engineering Lifecycle

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PHISHING TYPES

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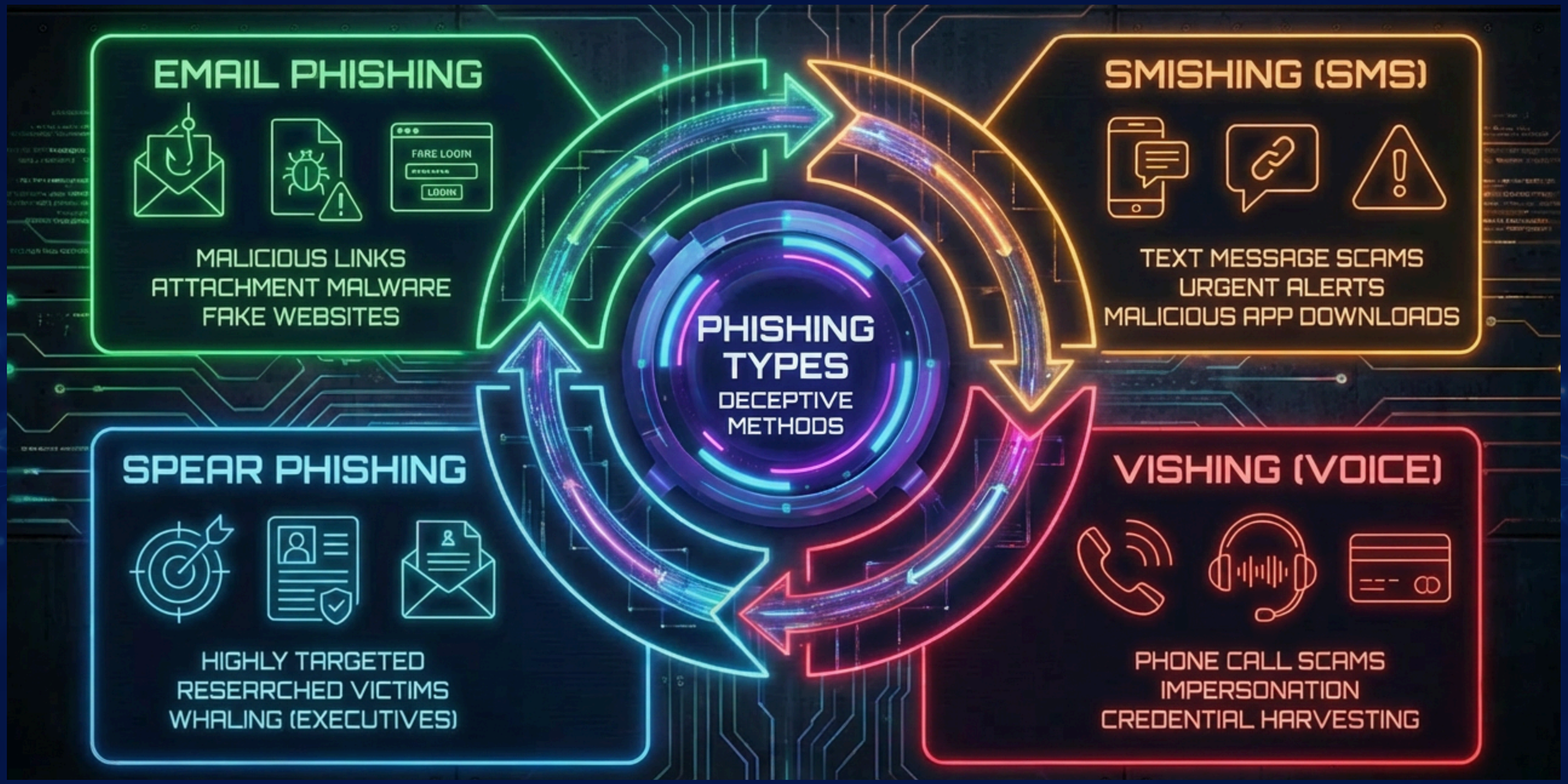

DEEPFAKE 101

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LLM 101

- Large Language Models (LLMs) are AI systems trained on massive datasets to generate human-like content.
- These models can write emails, create conversations, and mimic writing styles with remarkable accuracy.

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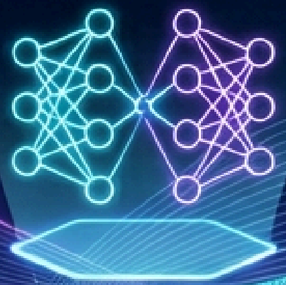
DEEPFAKE HISTORY



2014

GANs Introduced:
Ian Goodfellow publishes paper on Generative Adversarial Networks.

2017

First 'Deepfakes':
Reddit user shares AI-generated face-swaps, popularizing the term.

2019

Public Awareness & Detection:
Technology spreads; first major detection tools and legislative discussions begin.

PRESENT DAY

Advanced Realism & Threat:
High-fidelity real-time voice/video cloning. Used in sophisticated fraud and disinformation campaigns.

Generative Adversarial Networks

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Generator

The Generator creates synthetic content from random noise, attempting to produce outputs indistinguishable from real data. It learns to map latent space to realistic samples through iterative training.

Discriminator

The Discriminator acts as a critic, evaluating whether inputs are real or generated. It provides feedback that helps the Generator improve, creating an adversarial training loop.

Adversarial Loop

Both networks compete in a zero-sum game. As the Generator improves at creating fakes, the Discriminator gets better at detection, driving increasingly realistic outputs.

DEEPFAKE TYPES

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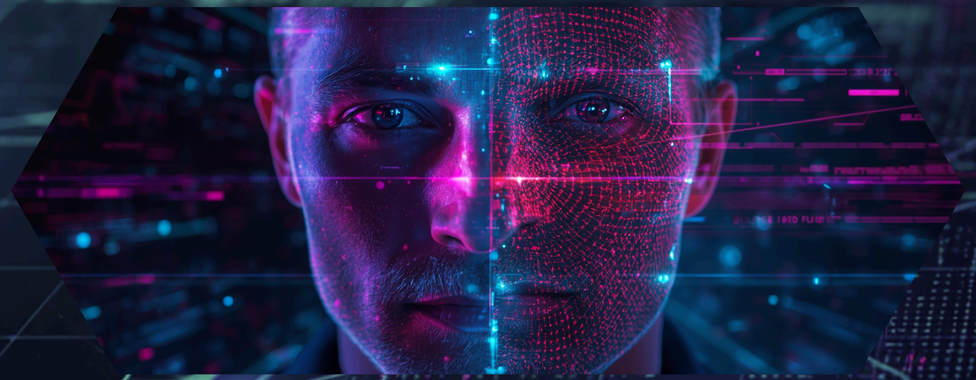

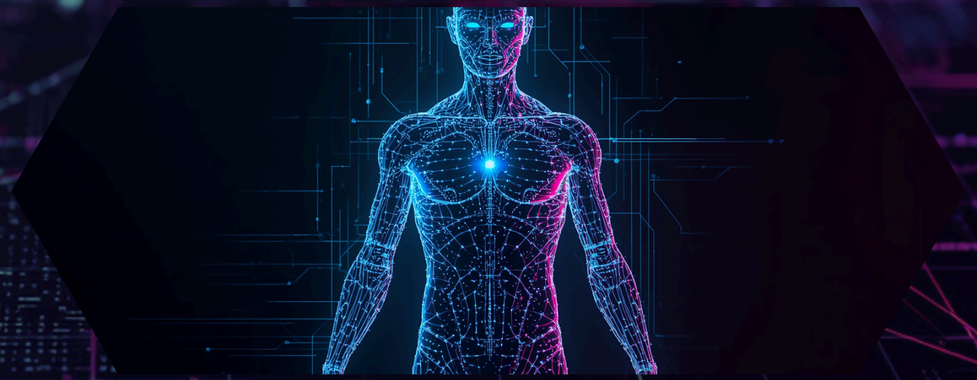

FACE SWAP

Replaces one person's face with another in video or images. Uses encoder-decoder neural networks to map facial features and expressions seamlessly onto a target face.

LIP SYNC

Manipulates mouth movements to match different audio. AI analyzes speech patterns and generates realistic lip movements synchronized with fabricated or altered audio tracks.

FULL BODY

Generates complete synthetic human figures with realistic movements and gestures. Combines pose estimation with generative models to create convincing full-body puppetry.

VOICE CLONE

Synthesizes realistic voice replicas from audio samples. Neural networks learn vocal characteristics, tone, and speech patterns to generate convincing synthetic speech.

AI-Powered Social Engineering

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Automated Targeting

AI algorithms analyze vast amounts of public data from social media, corporate websites, and leaked databases to identify high-value targets. Machine learning models profile victims based on behavior patterns, job roles, and network connections to maximize attack success rates.

Personalization

Large language models generate hyper-personalized phishing messages that reference specific projects, colleagues, and recent activities. These AI-crafted communications bypass traditional detection by mimicking authentic writing styles and contextual relevance.

Scale & Speed

AI enables attackers to launch thousands of unique, tailored attacks simultaneously. Automated systems adapt in real-time based on victim responses, creating dynamic conversation flows that maintain engagement and increase conversion rates dramatically.

GENERATE YOUR OWN DEEPFAKES (GYOD)

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Voice Cloning

1. OmniVoice (HuggingFace)
2. ElevenLabs (Paid)
3. Respeecher (Paid)

Video Deepfakes

1. DeepFaceLab
2. DeepFaceLive
3. FaceSwap

Case Study 01: Video Deepfake

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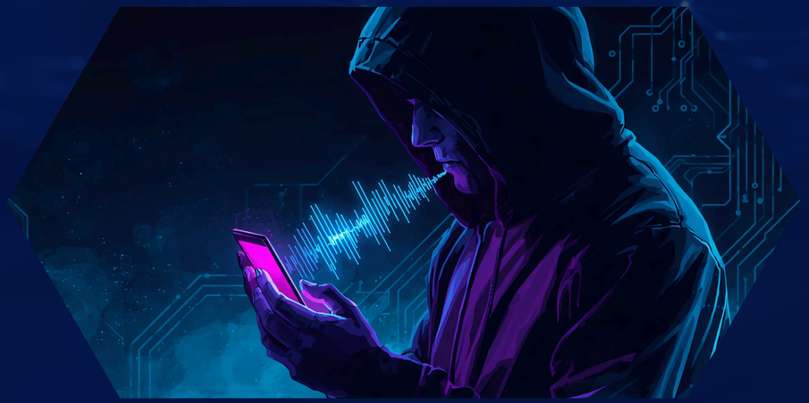
DEMO 01:

VIDEO

DEEPFAKE

Case Study 02: Voice Cloning



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DEMO 02:

VOICE CLONING

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Deepfake Detection Methods

Visual Artifacts

Look for unnatural blinking patterns, inconsistent lighting on faces, blurry edges around hairlines, and distorted facial features during movement. Pay attention to asymmetrical faces and misaligned teeth or ears.

Audio Analysis

Detect robotic or unnatural speech patterns, inconsistent background noise, and audio-visual sync issues. Voice cloning often produces subtle pitch anomalies and breathing pattern irregularities.

Metadata Review

Examine file metadata for editing software signatures, creation timestamps, and compression artifacts. Authentic media typically has consistent metadata trails while deepfakes show manipulation indicators.

AI Detection Tools

Leverage specialized AI tools which analyze pixel-level inconsistencies and neural network fingerprints invisible to human eyes.

Deepfake Detection Tools

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1. Awesome Deepfake Detection: https://github.com/Daisy-Zhang/Awesome-Deepfakes-Detection
2. Deep-Fake-Detector-v2-Model: https://huggingface.co/prithivMLmods/Deep-Fake-Detector-v2-Model
3. FaceForensics: https://github.com/ondyari/FaceForensics

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Key Takeaways

Awareness

Defense

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Q&A

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THANK YOU